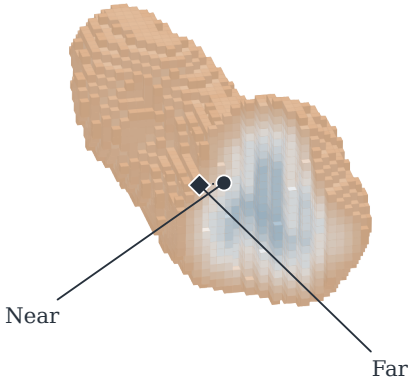
Same probability error, different distances

Two background voxels: reference $y = 0$, illustrative prediction $p = 0.25$



● Near

Distance value: 0.5 voxels

Penalty: $0.25 \times 0.5/8 = 0.015625$

◆ Far

Distance value: 3.5 voxels

Penalty: $0.25 \times 3.5/8 = 0.109375$

The farther voxel receives seven times the local penalty.



Normalized signed distance: blue inside; orange outside